

UMC 0.11um Logic and Mixed-Mode AI Advanced Enhancement Vertical NPN BJT Monte Carlo SPICE Model Document

**Version 0.1 Phase 1
2015/04/20**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
v0.1_p1	2015/04/20	Danny WC Lin	Original Release This Monte Carlo model including both process and mismatching characteristics are extracted from l110ae_bjt_npn_v011.mdl l110ae_bjt_npn_v011.lib

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al
Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

1.3 Model Usage

1.3.1 Simulators

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2013.3SP2
Cadence SPECTRE version 13.1.0.066 32bit
Mentor Graphics ELDO version 2013.1

1.3.2 The combination of statistical model & corner model

The Monte-Carlo model and corner model (3 corners) are combined into a single model library. User can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

MC: Monte-Carlo BJT model
TT: Typical BJT model
FF: Fast BJT model (Maximum beta model)
SS: Slow BJT model (Minimum beta model)

1.3.3 Mismatching parameters in corner model

The mismatching parameters are included in the corner models. Therefore, users can run Monte-Carlo mismatching simulation over a corner model. For example, users can evaluate the mismatching performance at the FF case.

1.3.4 Sigma number for Monte-Carlo model

The user can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. The Sigma=3 is default in the model and recommended.

1.3.5 Model usage

The definitions of sub-circuit model parameters

After invoking Monte-Carlo model parameters, the compact resistor models evolve to sub-circuit models. The input parameters for the sub-circuit model are the same as the compact model, except input parameter (mis_flag) are added for the sub-circuit model.

The following is the parameter list:

mis_flag: the flag parameter is used to turn on/off mismatching parameters for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.

Follow these steps to invoke the model file during simulation.

-Define the flag of Monte-Carlo simulation.

process: 1 to turn on the MC process simulation
 0 to turn off the MC process simulation

mismatch: 1 to turn on the local MC mismatch simulation
 0 to turn off the local MC mismatch simulation

Example:

-Define the number of Monte Carlo iterations.

Example:

sweep monte=1000 (for HSPICE and ELDO)

montecarlo numruns=1000 (for SPECTRE)

For Synopsys HSPICE

- Copy ModelFileName.lib into your simulation directory.

ModelFileName_MC.lib

ModelFileName_MC.mdl

ModelFileName_MC_CORNER.lib

ModelFileName_MC_STATISTICAL_P.mdl

ModelFileName_MC_STATISTICAL_M.mdl

- Add the following statement in the input file to the simulator.

.lib "/ ModelFileName_MC_CORNER.lib" CaseName

, where CaseName could be mc, tt, ff, ss.

- Define device condition by 'Qxx' statement.

Example:

X1 Vcc Vbb Vee DeviceName

For Cadence Spectre

- Copy ModelFileName.lib.scs into your simulation directory.

ModelFileName_MC.lib.scs

ModelFileName_MC.mdl.scs

ModelFileName_MC_CORNER.lib.scs

ModelFileName_MC_STATISTICAL_P.mdl.scs

ModelFileName_MC_STATISTICAL_M.mdl.scs

- Add the following statement in the input file to the simulator.

include "/ ModelFileName_MC_CORNER.lib.scs" section=CaseName

, where CaseName could be mc, tt, ff, ss.

- Define device condition by 'Qxx' statement.

Example:

X1 (Vcc Vbb Vee) DeviceName

For Mentor Graphics Eldo

- Copy ModelFileName.lib.eldo into your simulation directory.

ModelFileName_MC.lib.eldo

ModelFileName_MC.mdl.eldo

ModelFileName_MC_CORNER.lib.eldo

ModelFileName_MC_STATISTICAL_P.mdl.eldo

ModelFileName_MC_STATISTICAL_M.mdl.eldo

- Add the following statement in the input file to the simulator.

.lib "/ ModelFileName_MC_CORNER.lib.eldo" CaseName

, where CaseName could be mc, tt, ff, ss.

- Define device condition by 'Qxx' statement.

Example:

X1 Vcc Vbb Vee DeviceName

This model is valid within the following geometry and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry description:

Table 1-2 Vertical BJT device

Vertical NPN	nnpn_sv45x45_ L110AE	nnpn_sv90x90_ L110AE	nnpn_sv135x135_ L110AE
Emitter Area	4.5X4.5 μm^2	9X9 μm^2	13.5X13.5 μm^2
Base Area	11.7X11.7 μm^2	16.2X16.2 μm^2	20.7X20.7 μm^2
SAB (Salicide Block) overlap diffusion	Yes	Yes	Yes

Temperature range:

-40 C ~ +150 C

2 Monte Carlo DC Simulation Results for Process Variation

One thousand (1000) Monte Carlo iteration performed in the following simulation conditions, and the following corner points show TT, FF and SS values, individually

A reasonable number of Monte Carlo iteration is 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to (iteration times)^{-1/2}.

2.1 NPN Model Simulation Result Plots

2.1.1 NPN_SV45X45_L110AE

The following figures show the Monte Carlo simulation results of BJT's DC process variation at VBE=0.7V and VCE=3.3V of NPN_SV45X45_L110AE at 25 °C:

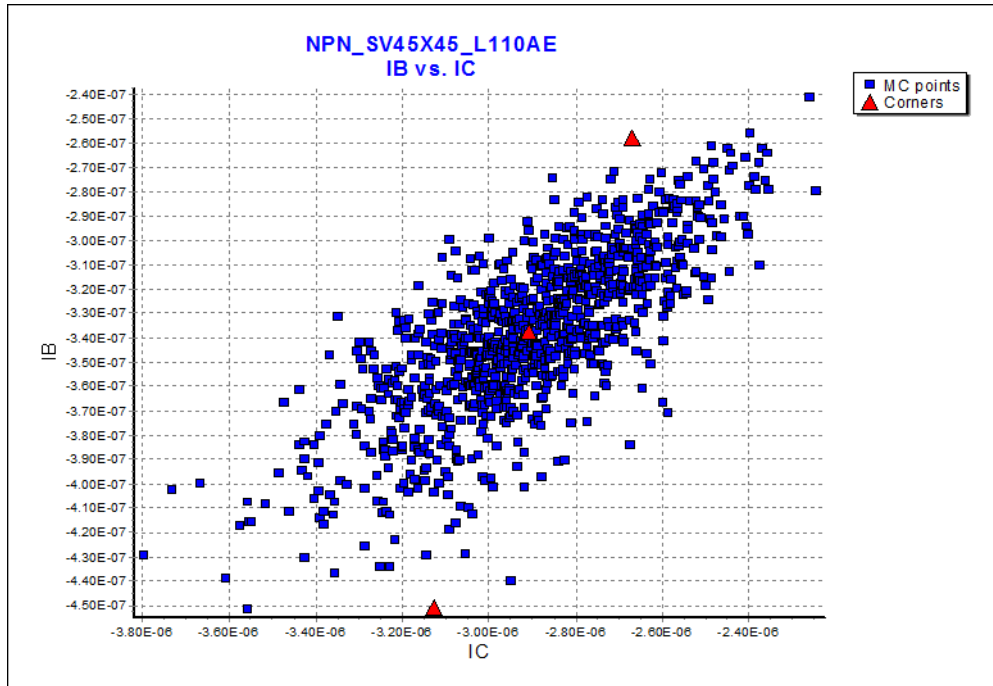


Figure 2-1 Monte Carlo plots of IB vs. IC of NPN_SV45X45_L110AE @ VBE=0.7V and VCE=3.3V

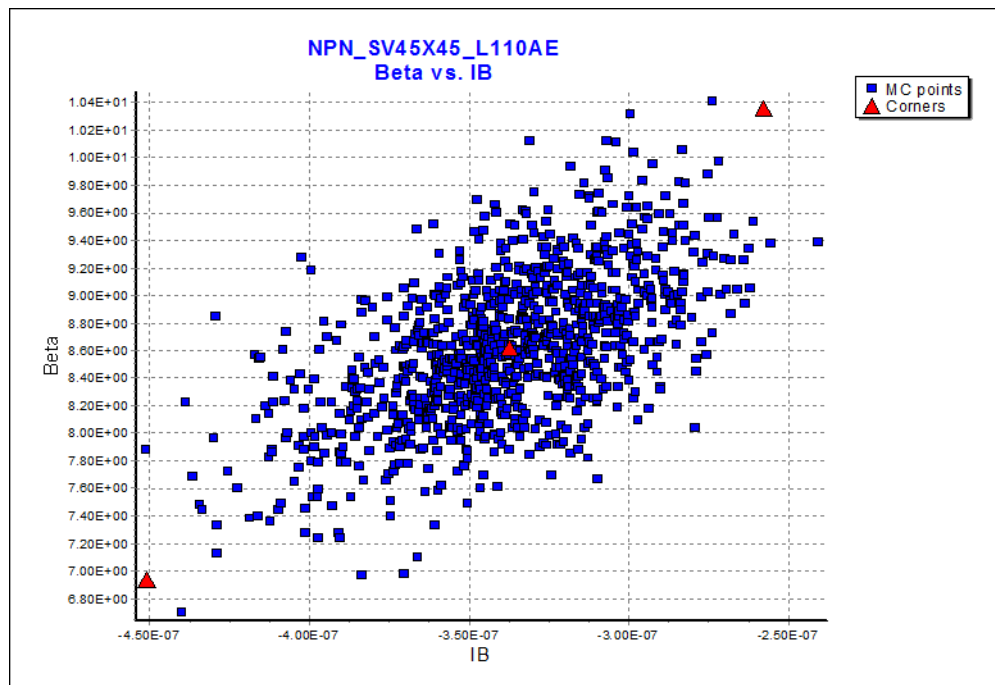


Figure 2-2 Monte Carlo plots of Beta vs. IB of NPN_SV45X45_L110AE @ VBE=0.7V and VCE=3.3V

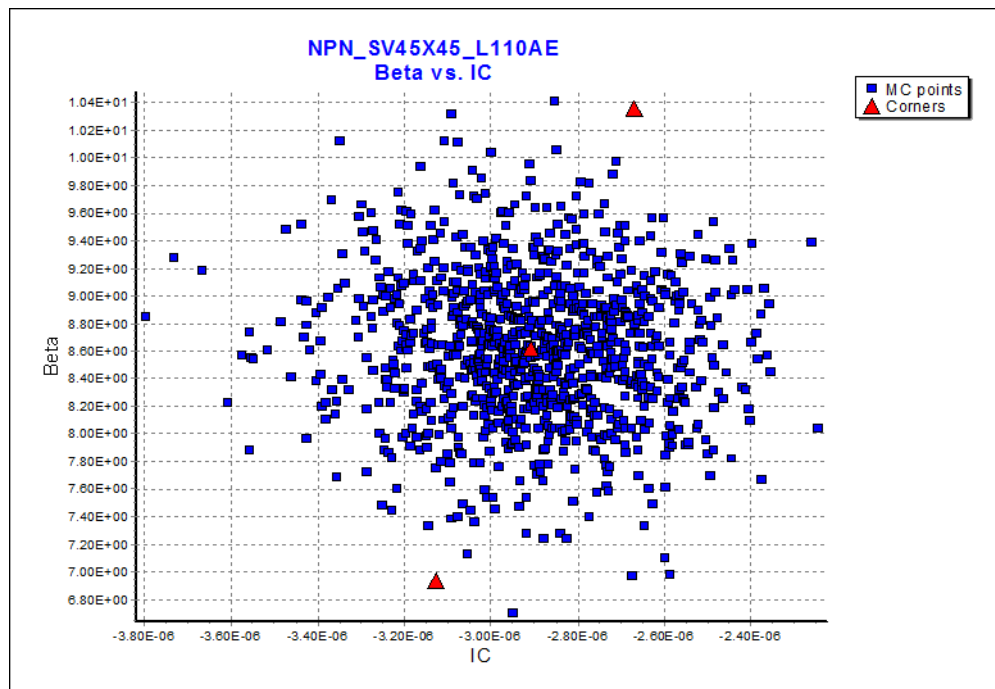


Figure 2-3 Monte Carlo plots of Beta vs. IC of NPN_SV45X45_L110AE @ VBE=0.7V and VCE=3.3V

2.1.2 NPN_SV90X90_L110AE

The following figures show the Monte Carlo simulation results of BJT's DC process variation at VBE=0.7V and VCE=3.3V of NPN_SV90X90_L110AE at 25 °C:

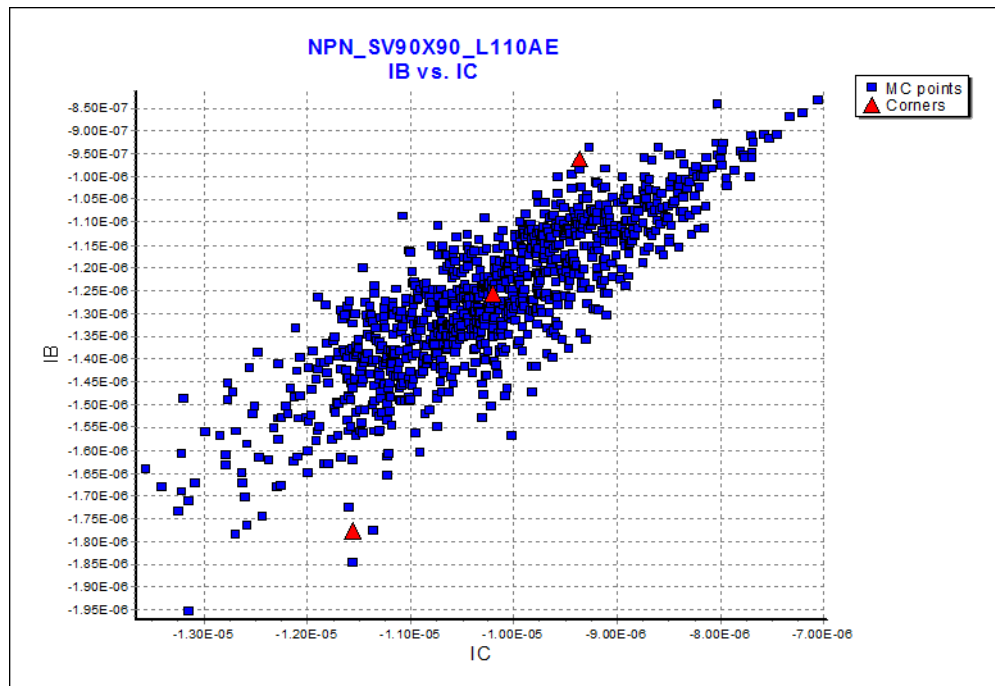


Figure 2-4 Monte Carlo plots of IB vs. IC of NPN_SV90X90_L110AE @ VBE=0.7V and VCE=3.3V

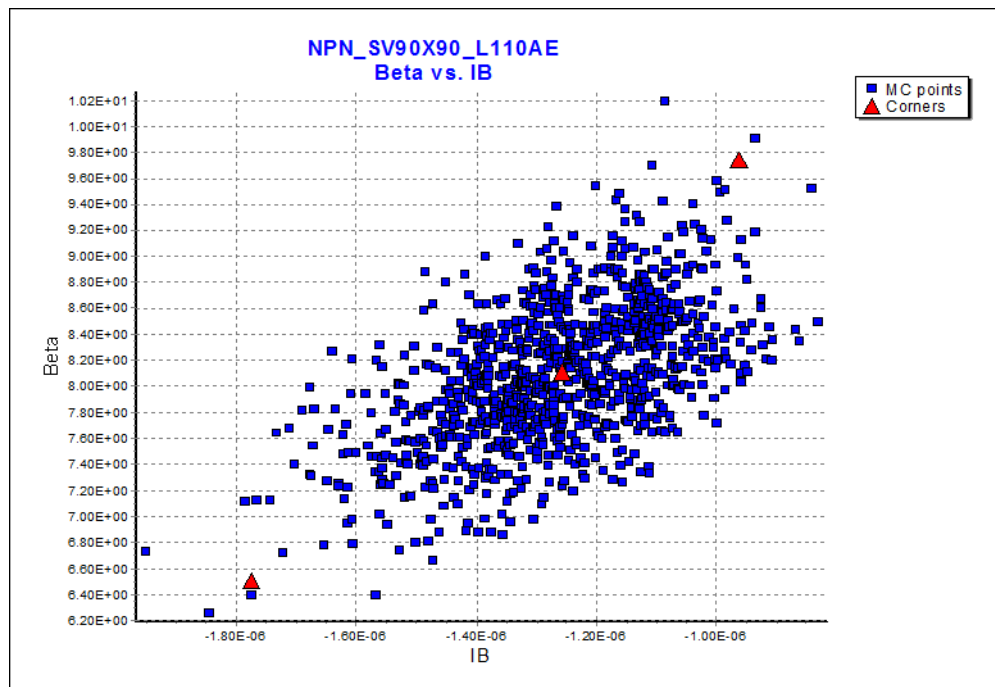


Figure 2-5 Monte Carlo plots of Beta vs. IB of NPN_SV90X90_L110AE @ VBE=0.7V and VCE=3.3V

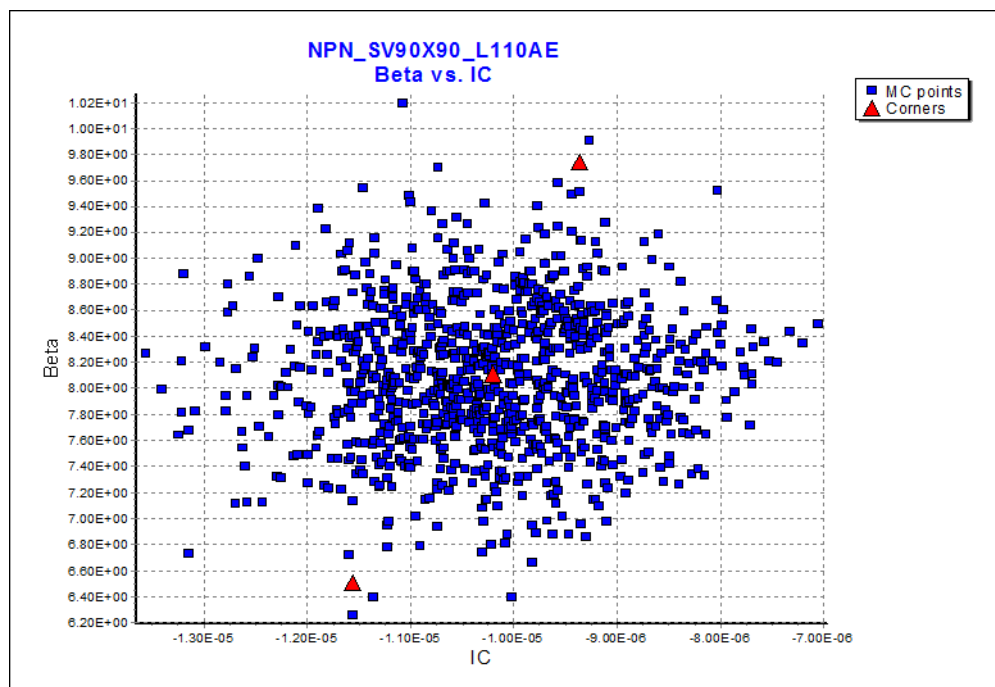


Figure 2-6 Monte Carlo plots of Beta vs. IC of NPN_SV90X90_L110AE @ VBE=0.7V and VCE=3.3V

2.1.3 NPN_SV135X135_L110AE

The following figures show the Monte Carlo simulation results of BJT's DC process variation at $V_{BE}=0.7V$ and $V_{CE}=3.3V$ of NPN_SV135X135_L110AE at $25^{\circ}C$:

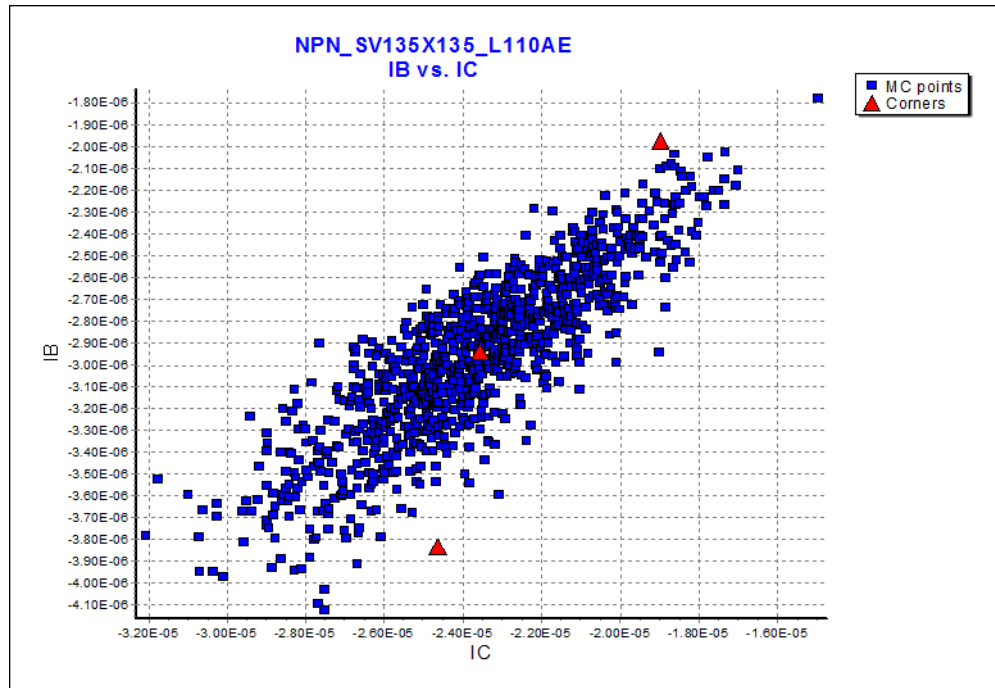


Figure 2-7 Monte Carlo plots of I_B vs. I_C of NPN_SV135X135_L110AE @ $V_{BE}=0.7V$ and $V_{CE}=3.3V$

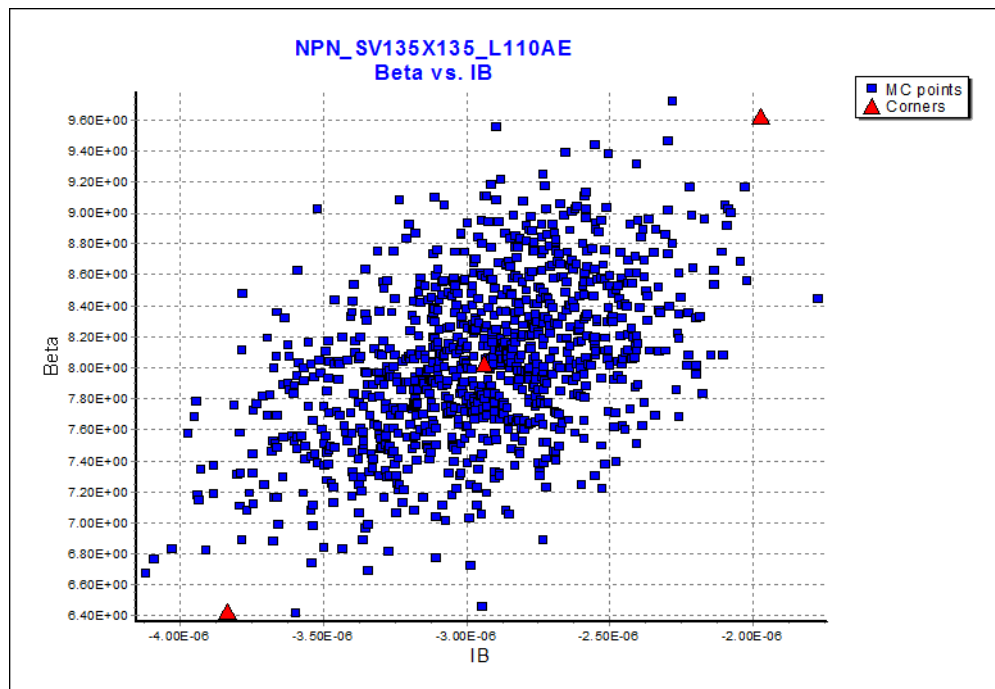


Figure 2-8 Monte Carlo plots of Beta vs. IB of NPN_SV135X135_L110AE @ VBE=0.7V and VCE=3.3V

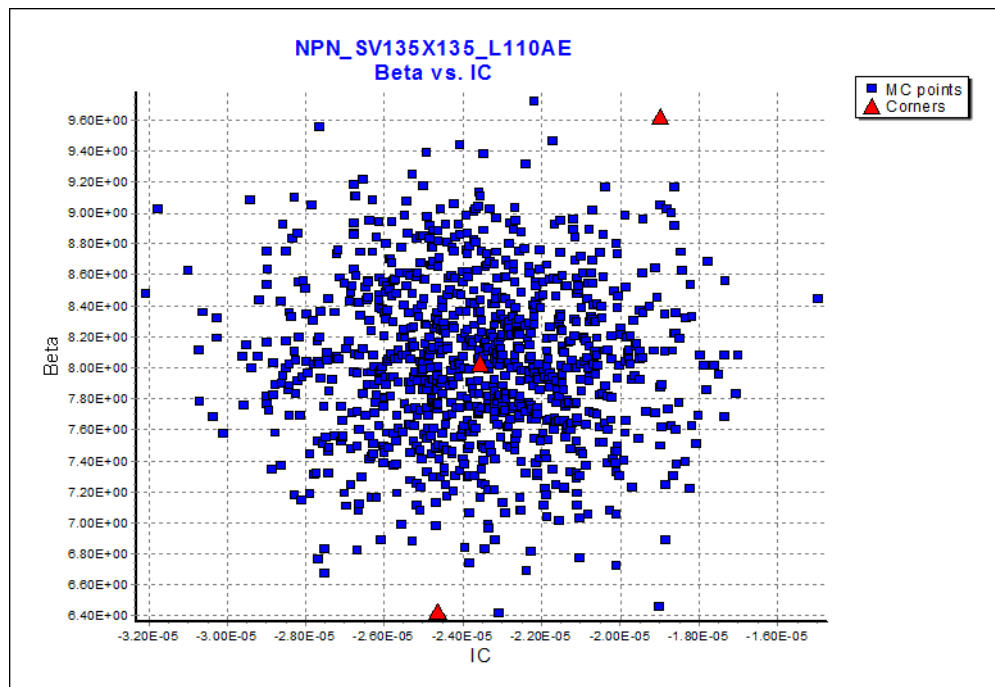


Figure 2-9 Monte Carlo plots of Beta vs. IC of NPN_SV135X135_L110AE @ VBE=0.7V and VCE=3.3V

3 Monte Carlo Simulation Results for Mismatch

3.1 NPN Simulation Result Plots

The following figures show the Monte Carlo mismatching simulation results at 25 °C.

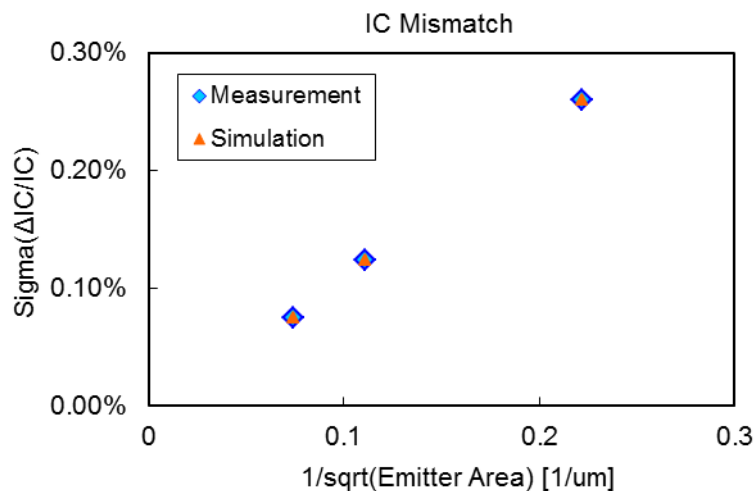


Figure 3-1 Mismatching simulation of IC for NPN

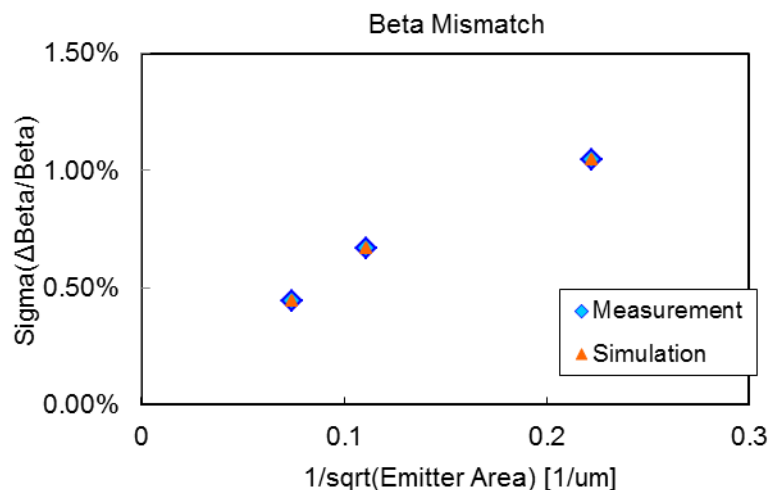


Figure 3-2 Mismatching simulation of Beta for NPN

4 Appendix

4.1 HSPICE Warning Message

Table 4-1 HSPICE warning message

NO.	Warning Message	Explanation
1	None	None

4.2 ELDO Warning Message

Table 4-2 ELDO warning message

NO.	Warning Message	Explanation
1	None	None

4.3 SPECTRE Warning Message

Table 4-3 SPECTRE warning message

NO.	Warning Message	Explanation
1	None	None